Inputs

- Observed unphased genotypes: Dad = 0/1, Mom = 0/1, Kid1 = 0/1, Kid2 = 0/1, Kid3 = 1/1.
- Inferred founder haplotype each kid carries: Kid1 = A|C, Kid2 = B|D, Kid3 = A|D.

Algorithm

Enumerate the 4 ways to assign alleles (0 and 1) to dad's two haplotypes A, B and mom's two haplotypes C, D consistent with the parents' unphased genotypes. For each such assignment:

1. derive each kid's expected phased genotype by reading off the alleles assigned to the two founder haplotypes it carries (e.g., A|C → dad's A allele assignment paired with mom's C allele assignment);
2. for each kid, unphase the expected genotype and tick (✓) if it matches the kid's observed unphased genotype (from Inputs);
3. count kids whose expected unphased genotype agrees with observed (# Matches).

Hap-allele assignment				Expected kid genotypes						
Dad (0/1)		Mom (0/1)		Kid1		Kid2		Kid3		# Matches
A	B	C	D	A|C		B|D		A|D		
0	1	0	1	0|0	0/0	1|1	1/1	0|1	0/1	0
0	1	1	0	0|1	0/1 ✔	1|0	0/1 ✔	0|0	0/0	2
1	0	0	1	1|0	0/1 ✔	0|1	0/1 ✔	1|1	1/1 ✔	3
1	0	1	0	1|1	1/1	0|0	0/0	1|0	0/1	0

Output

The assignment with the most matches (boxed) gives the deduced phased genotypes:

Kid1 = 1|0, Kid2 = 0|1, Kid3 = 1|1.

Reconstructed haplotype sequences in IBD segments

Kid1 pat_A:	1	1	0	1	1
Kid1 mat_C:	1	0	1	1	0
Kid2 pat_B:	0	0	0	0	0
Kid2 mat_D:	1	1	0	0	0
Kid3 pat_A:	1	1	0	1	1
Kid3 mat_D:	1	1	0	0	0
	●	○	●	○	●

● Informative (see Fig 2)

○ Non-informative (this figure)